

Problem 2.

(a)

the truth table shows that the $(P \wedge \neg Q) \vee (P \wedge Q)$ is equivalent to P .

P	Q	$\neg Q$	$P \wedge \neg Q$	$P \wedge Q$	$(P \wedge \neg Q) \vee (P \wedge Q)$
T	T	F	F	T	T
T	F	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F

(b)

$(A - B)$ is equivalent to $A \wedge \neg B$

the truth table about $(A - B) \vee (A \wedge B)$ is

A	B	$\neg B$	$A \wedge \neg B$	$A \wedge B$	$(A - B) \vee (A \wedge B)$
T	T	F	F	T	T
T	F	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F

we can see the A is equivalent to $(A - B) \vee (A \wedge B)$

so $x \in A \iff x \in ((A - B) \vee (A \wedge B))$

Problem 3.

(a) $\text{sixish}(n) := \exists m \exists k. ((6 \cdot 10^k + m = n) \wedge m < 10^k)$

(b) $\exists m \exists n \forall k. Q(m) \wedge Q(n) \wedge k \neq m \wedge k \neq n \wedge \neg Q(k)$

(c) $\forall k \exists n \exists m. (\text{isEven}(k) \wedge k > 2) \implies (m + n = k) \wedge \text{isPrime}(n) \wedge \text{isPrime}(m)$

(d) $\forall n. n > 100 \implies \text{sixish}(n)$

(e) it is not equivalent because when $n = 1$, $R(n)$ should be True but we can't find any $b \in \mathbb{N}$ to make $1 = 2b + 5$

Problem 4.